

Follow-up Observations of p-mode Pulsating Hot Subdwarf TIC 63168679 (2M0753+2324)

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ABSTRACT

This study presents ground-based follow-up observations of the p-mode pulsating hot subdwarf TIC 63168679 (2M0753+2324) to verify pulsation frequencies previously identified by the Transiting Exoplanet Survey Satellite (TESS). Utilizing the Steward Observatory 61-inch Kuiper Telescope with its Mont4K CCD imager, we obtained 806 images over a six-hour period using a Schott 8612 filter and an 18-second exposure cadence. Through aperture photometry and a Nyquist-limited Lomb-Scargle periodogram analysis, five distinct pulsation frequencies were detected, including a primary signal at $7130.16\mu\text{Hz}$ that closely matches past literature. The results estimate the star’s physical parameters at an effective temperature of approximately 35,000 K and a surface gravity of $\log g \approx 5.8$, identifying it as a terminal-age extreme horizontal branch (TAEHB) star with a core mass of $0.47M_{\odot}$. Future work involves continuous photometric monitoring of this object in order to resolve the possible pulsation multiplets, which can help us derive the star’s rotational period.

1. INTRODUCTION

Hot subdwarf stars are evolved, core helium-burning stars with spectral types from O to B, making them hotter and brighter than typical subdwarf stars. The B-type subdwarfs (sdBs) have surface gravities $5 \leq \log(g/\text{cm s}^{-2}) \leq 6$ and effective temperatures T_{eff} between 20,000 and 40,000 K, placing them at the hotter, blue end of the horizontal branch, which is also referred to as the extreme horizontal branch (EHB) (Heber 1986; Saffer et al. 1994). They are anticipated to evolve from intermediate-mass stars with around $0.7\text{--}2.2 M_{\odot}$ that experienced significant mass loss, leaving behind a tiny portion of their original hydrogen envelopes (Baran et al. 2023). A variety of theories for their formation have been proposed and modeled, from single star evolution scenario with enhanced mass loss at the tip of the red giant branch (D’Cruz et al. 1996) to common envelope evolution, Roche lobe overflow, and mergers in binary evolution (Mengel et al. 1976; Han et al. 2002, 2003).

Meanwhile, O-type subdwarfs (sdOs) are hotter ($T_{\text{eff}} = 20\,000\text{--}40\,000$ K) and have a wider range of surface gravities ($\log(g/\text{cm s}^{-2}) = 4.0\text{--}6.5$), with He-enriched atmospheres (Baran et al. 2023). There are many diverse proposed scenarios about the origins of these sdOs. While Miller Bertolami et al. (2008) suggest that the majority of the more compact sdOs are formed after late hot helium flashes, Webbink (1984) and Saio & Jeffery (2000, 2002) associate them with merger events. Moreover, Reindl et al. (2016) conclude that the less compact ones with $\log(g/\text{cm s}^{-2}) \lesssim 5.0$ have proceeded with the traditional evolutionary scheme of post-AGB stars by climbing the asymptotic giant branch after running out of core helium for nuclear fusion.

Independently discovered by Kilkeny et al. (1997) and Charpinet et al. (1996, 1997), stellar oscillations in hot pulsators like sdBs provide a useful way to estimate their masses precisely and probe information about their interior structures with asteroseismology. Kilkeny et al. (1997) reported pressure (p-) mode pulsators in this paper and subsequent work in their series, while Green et al. (2003), Baran et al. (2005), and Schuh et al. (2006) were able to find gravity (g-) mode sdBs and hybrid ones that exhibit both p- and g-modes, all of which are driven by the κ -mechanism, powered by a sharp increase in opacity due to local enhancements of iron abundance in the sdBs’ envelopes Charpinet et al. (1997). Radiation pressure pushes iron and heavy elements towards the outer layers of the hot envelope, causing them to be ionized and thus a sharp increase in opacity. The higher opacity traps radiation inside the compressed layer, increasing the local pressure and driving the layer further outward. As this layer expands, it becomes cooler and less opaque, enabling the trapped heat to escape. Gravity of the star then pulls the layer back inward, and the cycle repeats, resulting in the observed, periodic pulsations. Additionally, Fontaine et al. (2012) note that p-mode pulsators typically have shorter periods, while amplitudes of the pulsations of g-mode sdBs are higher.

More information about sdBs and sdOs can be found in Baran et al. (2023, hereafter, Paper I). In the search for p-mode hot subdwarfs in both northern and southern ecliptic hemispheres by the Transiting Exoplanet Survey Satellite (TESS) space telescope between year 1 and year 4 of the mission, featured by Paper I and Baran et al. (2024, hereafter, Paper II), a total of 93 short-period hot-subdwarf pulsators (43 in Paper I and 50 in Paper II) is clearly identified. The target of this paper, TIC 63168679 (2M0753+2324, shorthand for 2MASS J07532911+2324363), was observed in the northern ecliptic hemisphere survey and reported in Paper II. TIC 63168679 is a recently discovered intermediate He-rich hot subdwarfs (sdOB) in Lei et al. (2020). It is one of the few sub-dwarfs with intermediate He abundance ($-1 \leq \log(\frac{n_{He}}{n_H}) \leq 0$), and strong enrichment in heavy metals, such as zirconium and lead, as described in Paper I. Paper II reports ten frequencies of this pulsator between 6657 and 7340 μHz with possible multiplets, especially an asymmetric triplet around 6657 μHz that they needed better resolution to confirm.

The goal of this paper is to confirm the frequencies of the pulsations reported in Paper II using ground-based observations with similar cadences to the ones used for TESS observations. Furthermore, despite the worse observing conditions due to atmospheric turbulence and lower resolution compared to TESS, we will attempt to provide insights into the confidence about the possible pulsation triplet mentioned in Paper II. And as discussed previously, the resulting light curve that we obtained can also be used to model for pulsation modes and probe for information about the star's interior structure using asteroseismology. Now that we have discussed relevant scientific context about our object, we will now discuss our photometric data was acquired and reduced to extract the target's pulsating frequencies in section 2, showcase our results in section 3, and address the importance of our results and their implications in section 4, before summarizing what we have done in section 5.

2. OBSERVATIONS & DATA ANALYSIS

2.1. Telescope & Instrument

The Steward Observatory 61-inch Kuiper Telescope, located at an elevation of 2510 m (8235 ft) on Mount Bigelow, Tucson, AZ, has a primary mirror diameter of 1.54 m (61 inches) and a primary focal ratio of f/4. Because we observed at visible wavelengths, we used the telescope's f/13.5 Cassegrain secondary with a diameter of 40.96 cm, plate scale of 10.0 arcsec/mm, and useful field of view (FOV) of >435 arcsec diameter. The Mont4K (as in Montréal 4K $\times$ 4K CCD Imager) has a size of 4096 $\times$ 4096 pixels and image scale of 0.14 arcsec/pixel, resulting in a FOV of 580 $\times$ 580 arcsec². The Mont4K has a response of more than 95% in the blue band, making it ideal for analysing the pulsations of hot subdwarfs such as TIC 63168679 (Fontaine et al. 2014). More information about the telescope is available at the following web address, <https://james.as.arizona.edu/~psmith/61inch/>; and more details about the Mont4K can be found at the following website, <https://james.as.arizona.edu/~psmith/61inch/CCD/CCDmanual.html>. Since the target is brightest in the Schott 8612 filter, which offers a bandpass of 3000-7000 Å with a magnified 4" $\times$ 4" FOV while cutting off near-IR signals completely (see Fig. 1), the observations were done using this filter.

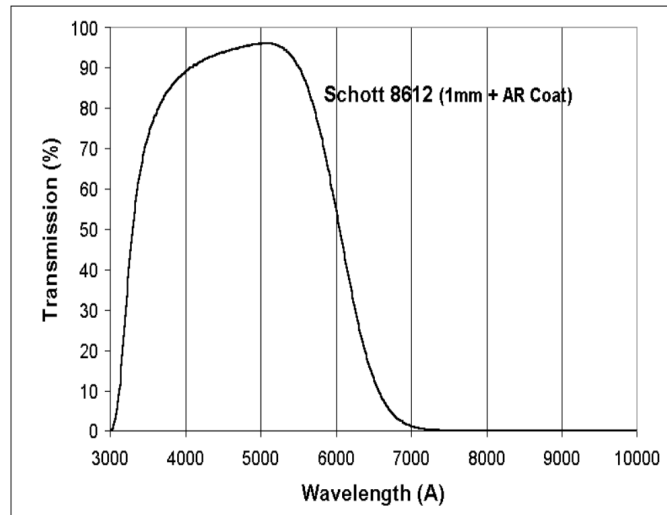


Figure 1: S-filter bandpass tracing, per the [Mont4K CCD Operating Manual](#).

2.2. Data Acquisition

We observed TIC 63168679 with the Steward Observatory 61-inch Kuiper Telescope Mont4K CCD Imager on the night of 2026 March 6-7 (UTC -7). Information about the observed target is available in Table 1. The observations were done under the supervision of Dr. E. M. Green, who also kindly provided us with a finder chart showing the target and all reference stars used to derive the relative flux (see Fig. 15).

Object Name (Paper II)	Other Designation	RA (J2000)	Declination (J2000)	V	Object Type
TIC 63168679	2M0753+2324	07:53:29.1	+23:24:36	13.27	sdOB pulsator

Table 1: Information about TIC 63168679. The right ascension and declination of the target are queried from SIMBAD (Gaia Collaboration 2020), which are not the precise coordinates but can be used as a starting point to locate its true location at the beginning of the observation.

Over the course of ~ 6 hours from dusk ($\sim 19:11$) of 2026 March 6 (UTC -7) to when the target’s altitude was below the safety limits of the facility at $\sim 01:45$ of 2026 March 7 (UTC -7), we took a total of 806 images of our target with an exposure time of 18 seconds each, which is relatively close to the 20s integration time used for our target as described in Paper II. Seeing was good throughout the observing session, improving from $1.8''$ - $2.1''$ at the beginning to $1.6''$ later on, with sometimes being less than $1.3''$; however, as the night progressed, there was more and more cloud coverage. For each exposure, using scripts by Dr. Green on the telescope’s computer, the extended FITS images were combined into one normal FITS image with overscan subtraction and interpolation across bad columns of the CCD was also done (see Fig. 15 for the locations of the bad columns). Apart from that, we took bias frames to account for the high electron read noise (5 electrons) of the Mont4K, and flat frames to account for vignette in each image, which are noticeable in the corners of the images when $4'' \times 4''$ when the S-filter is used, per the [Mont4K CCD Operating Manual](#). Even though the Mont4K has a high dark current rate (16.6 electrons/pixel/hour), dark frames were not taken since the S-filter already has a high steepness of near-IR cutoff. The bias frames were combined into a single superbias image by Dr. Green; and the flat frames were also combined into a single flat image, which was then normalized by subtracting the combined bias frame from it and dividing the differences by the median of the differences to give the normalized superflat image.

2.3. Calibrations

First, we calibrated individual frames to remove measurement artifacts using the standard image analysis routine of sky background subtraction, flat fielding, and bad pixel correction (bias subtraction). For each raw science image, we calculated the median of the pixel values of three 100×100 rectangular regions that are free of stars, and calculated the median of these three values to determine the sky background value. After that, we reduced the science image by subtracting the superbias frame from the raw science frame, dividing the difference by the normalized superflat frame, and subtracting the sky background value from the result. After this first stage of data processing, a comparison between a single raw image and that same image after being reduced is available in Fig. 5.

2.4. Aperture Photometry

After calibrating all the individual frames, we performed aperture photometry using the python module `photutils` from `Astropy` to extract the flux and flux errors of the target and all 11 reference stars (see Fig. 3 for their approximate positions in an example image and Table 2 for their approximate pixel coordinates in all the reduced images provided by Dr. Green). For each star (either target or reference star), we created a circular aperture of radius $2.25 \times$ its average full width at half maximum (FWHM) around its specified location, which would give us the most precise result as in Dr. Green’s instructions. We then created a circular annulus around the same position with an inner radius of 13 pixels, since the image with the worst seeing is measured well with an aperture radius of 11 pixels, and an outer radius of 18 pixels to avoid contamination from nearby stars. Later on, we were informed by Dr. Green that the stars drifted noticeably throughout the observations (see Fig. 4 for more details). However, since the radii were always larger than the total drifts of the stars (see Fig. 5b), our circular aperture would always surround the star, thus the flux would always be computed accurately.

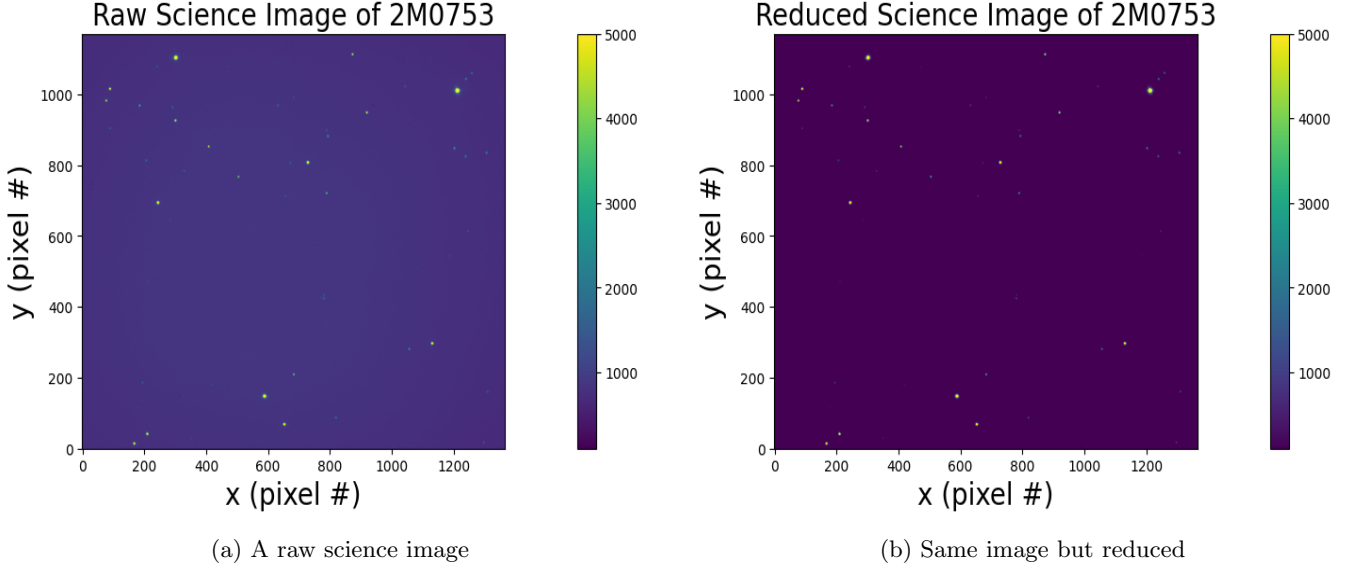


Figure 2: A comparison between a raw science image and that same image after being reduced with our routine.

Star #	x (pixel #)	y (pixel #)
1	653.16	70.03
2	1130.54	298.10
3	245.35	694.94
4	783.93	721.77
5	729.00	808.00
6	1306.31	835.58
7	1202.37	848.19
8	409.20	853.11
9	301.97	926.50
10	919.89	949.05
11	90.50	1015.82
12	873.49	1113.11

Table 2: The approximate pixel coordinates of TIC 63168679 (#5, written in bold) and all the 11 reference stars that we used to derive the target’s relative flux

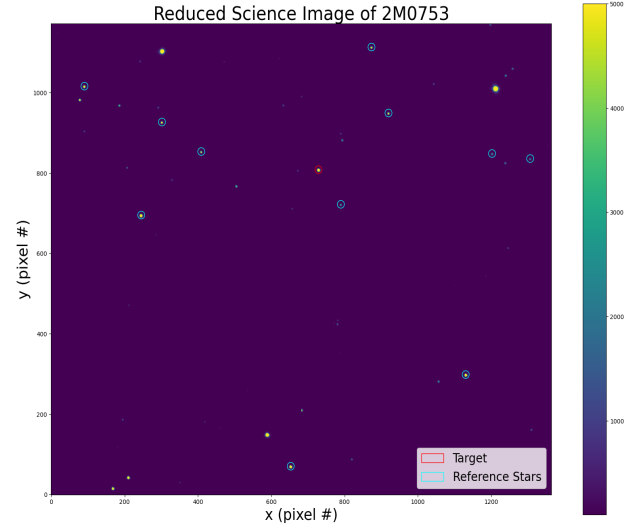


Figure 3: An example reduced science image with markers of where TIC 63168679 and all reference stars are

With the circular aperture and annulus, we used photutils’ built-in function `aperture.aperture_photometry` to extract the flux within the area, and the uncertainty in flux is

$$\sigma_{Flux} = \sqrt{A_{apt}} * \sigma_{bkg_annulus} \quad (1)$$

where σ_{Flux} is the uncertainty of the extracted flux, A_{apt} is the area of the circular aperture, and $\sigma_{bkg_annulus}$ is the standard deviation of the sky background within the annulus, which can be extracted using photutils’ built-in function `aperture.ApertureStats`. The same thing was done to extract the fluxes and flux uncertainties of all the stars in one image. This aperture photometry routine was used across all 806 reduced images.

After that, we normalized the flux of TIC 63168679 as follows,

$$normalized_flux = \frac{raw_flux - phot_corr}{mean(raw_flux - phot_corr)} \quad (2)$$

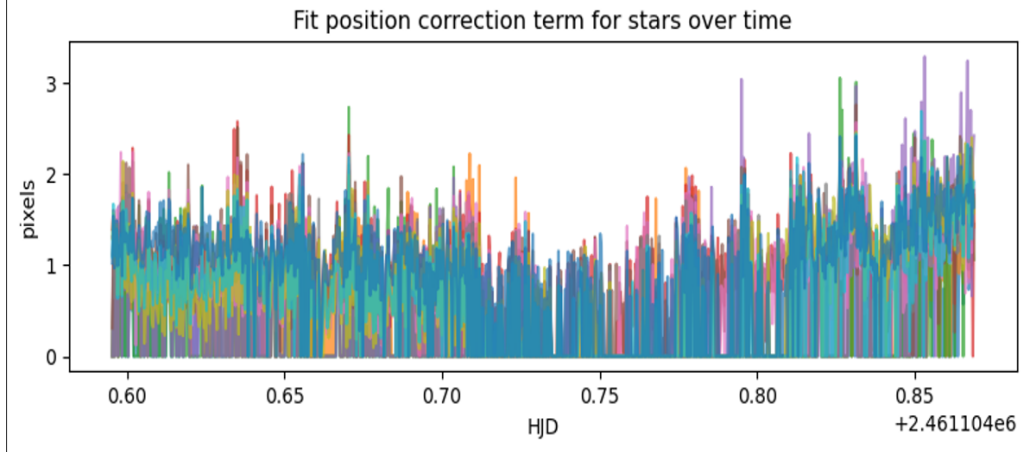
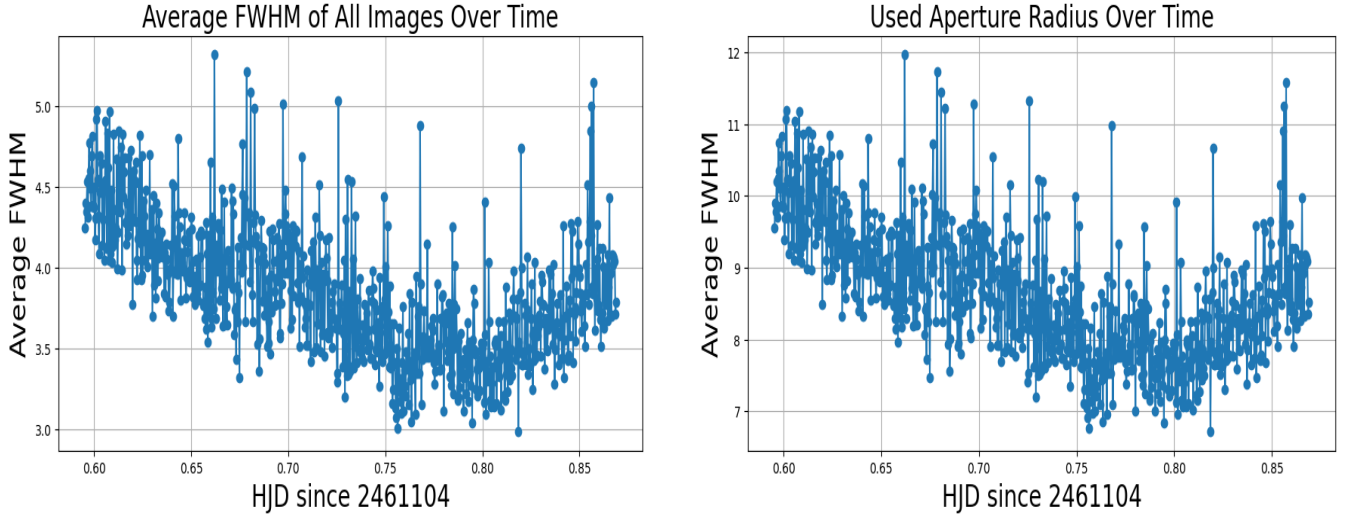


Figure 4: Total drift $\sqrt{(\Delta x)^2 + (\Delta y)^2}$ in the positions of target & reference stars over time.



(a) Average FWHM of target and all reference stars over Time (b) Radius of the Circular Aperature over Time ($2.25 \times \text{FWHM}$)

Figure 5: The average FWHM of the target & all reference stars, and the used radius of the aperture over time.

where raw_flux is the extracted flux of TIC 63168679 from the aperture photometry routine, $phot_corr$ is the photometric correction factor, which is defined as the mean of the fluxes of the reference stars. Therefore, the two sources of errors of the normalized flux come from the uncertainty in extracted flux of the target and the uncertainty in the so-called photometric correction factor. The total uncertainty in the normalized flux can then be defined as follows,

$$\sigma_{normalized_flux} = \frac{\sqrt{(\sigma_{raw_flux})^2 + (\sigma_{raw_flux} \sigma_{phot_corr})^2}}{\sigma_{phot_corr}} \quad (3)$$

where $\sigma_{normalized_flux}$ is the total error of the normalized flux, σ_{raw_flux} is the uncertainty in the raw flux, σ_{phot_corr} is the uncertainty in the photometric correction factor, which comes from the errors in the fluxes of reference stars that are used to calculate the mean and can be defined as follows,

$$\sigma_{phot_corr} = \frac{\sqrt{\sigma_{ref1}^2 + \sigma_{ref2}^2 + \sigma_{ref3}^2 + \dots + \sigma_{ref11}^2}}{11} \quad (4)$$

where σ_{ref1} is uncertainty in flux of reference star #1, σ_{ref2} is uncertainty in flux of reference star #2, σ_{ref3} is uncertainty in flux of reference star #3, and so on, and there are 11 reference stars in total (see Table. 2).

2.5. Lomb-Scargle Periodogram

Finally, after obtaining the normalized flux from the previous step, we performed least-squares analysis, particularly the Lomb-Scargle (LS) periodogram, to estimate the frequency spectrum by fitting sinusoidal functions to our photometric data. This can be done by using *Astropy*'s eponymous function in its module *timeseries*. Alternatively, we could have sent our data (time stamps and normalized flux values) to Dr. Green, who could help us interpolate the data points to obtain data points with evenly-spaced time values that can be inputted to a fast Fourier transform (FFT) routine. However, since LS can deal with our unevenly-spaced data already and identify low-frequency periodic signals while FFT interpolation can reduce accuracy, we decided to proceed with the LS method.

First, we converted the time values (in heliocentric Julian day or HJD that would be inputted to our LS routine) by dividing them by 86400 to convert to Hz and multiplying by 10^6 to convert to μHz , since the expected frequencies of the pulsations from Paper II are in this unit. After that, we inputted the (converted) time values, normalized flux values, and uncertainties in normalized value to the mentioned function to obtain the LS periodogram. As noted earlier, the integration time of our data is similar to the USC cadence described in Paper II. Hence, we also adopted a frequency upper-limit at the Nyquist frequency, which is defined as $\frac{1}{2 \times \text{median}(\text{frequency})}$ (see [VanderPlas \(2018\)](#) for details about this so-called Nyquist factor of the LS periodogram), in order to sample the entire p-mode region. Moreover, although photometric data that is irregularly sampled can extract frequencies beyond the expected Nyquist frequency for an equivalent FFT, the resulting frequencies are subject to more poorly-defined errors and aliasing, resulting in less confidence of detection.

Similar to Paper II, we also adopted a detection threshold of $5.5 \times \text{median noise level}$, corresponding to a false alarm probability (FAP) of 0.1%. Thus, any frequencies with $S/N \leq 5.5$ would be considered tentative (see Paper II and [Baran & Koen \(2021\)](#) for more precise confidence level derivations). In order to search for significant signals in the frequency spectrum, we conducted an automatic search for any flux variations beyond 100 μHz and then checked if such variations are higher than the defined FAP level. The amplitudes of the pulsations are then the amplitudes of the power peaks, outputted by the (Nyquist-limited) LS routine.

3. RESULTS

3.1. Signal-to-Noise Ratio

Since the observations only spanned ~ 6 hours on the nights of March 6-7, 2026 (HJD = 2461104-2461105), we subtracted the HJD of all images by 2461104 so that the time axis looks neat. Then, we calculated the signal-to-noise ratio (SNR) of TIC 63168679 as follows, $S/N = \frac{Flux}{\sigma_{Flux}}$, where $Flux$ is the flux values of the star (can be either normalized or unnormalized) and σ_{Flux} is the uncertainty in unnormalized flux values in Eq. 1 or in normalized flux values in Eq. 2. The results for unnormalized and normalized fluxes are shown in Fig. 6a and Fig. 6b, respectively. The SNR decreased as the observations progressed due to the increasing cloud coverage, but we managed to obtain photometric data with $SNR > 1500$ over the entire session. Moreover, the SNR was relatively high at the beginning because we started observing at around dusk, when the sky was still bright. Therefore, the CCD picked up excessive sky signals that were not removed entirely by sky background subtraction in our calibration routine. In addition, since the SNR is the same for both unnormalized and normalized fluxes, it implies that we normalized the flux and estimated the corresponding errors accurately.

3.2. Light Curves

Similar modifications to the time values, as described in section 3.1, were also done to the time stamps of the light curves. We extracted the fluxes of TIC 63168679 & all reference stars and estimated the uncertainties in flux for all of them. The resulting light curve for all the objects (TIC 63168679 & reference stars) is Fig. 7b, while Fig. 7a is the light curve of our target only. The decrease in the count of TIC 63168679 and some reference stars can also be explained by the increase in cloud coverage over the course of the observing session, similar to the drop in SNR in Fig. 6. And those with flat light curves are simply because there were no clouds passing in front of them for the duration of the observations.

In addition, we normalized the flux of TIC 63168679 with fluxes of all reference stars and estimated the uncertainties in the normalized flux, as described in section 2.4, and the result is Fig. 8. The drop in the count in the full light curve here can also be explained by cloud coverage over the target and some reference stars, as in Fig. 7. While the peaks appeared to be random noise from measurements in the full light curve (see 8a, when we zoomed in on a portion, these little peaks are more than 5σ , thus they cannot be explained by random noise and must then be pulsations of

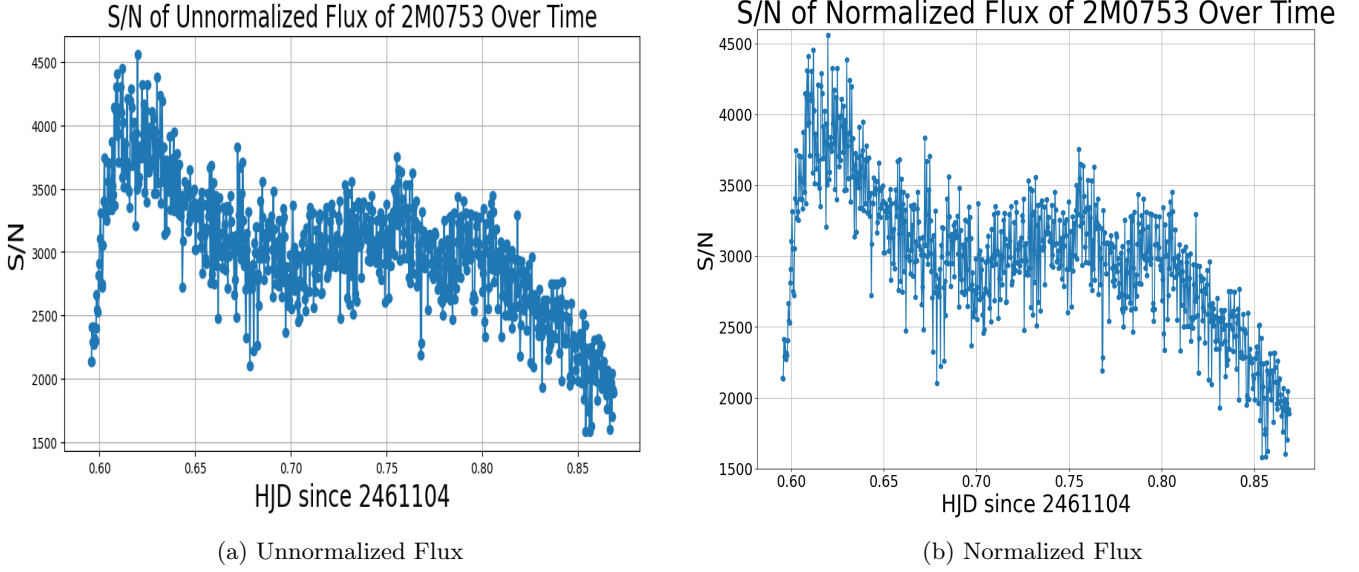


Figure 6: S/N Ratio of the Unnormalized and Normalized Flux of TIC 63168679

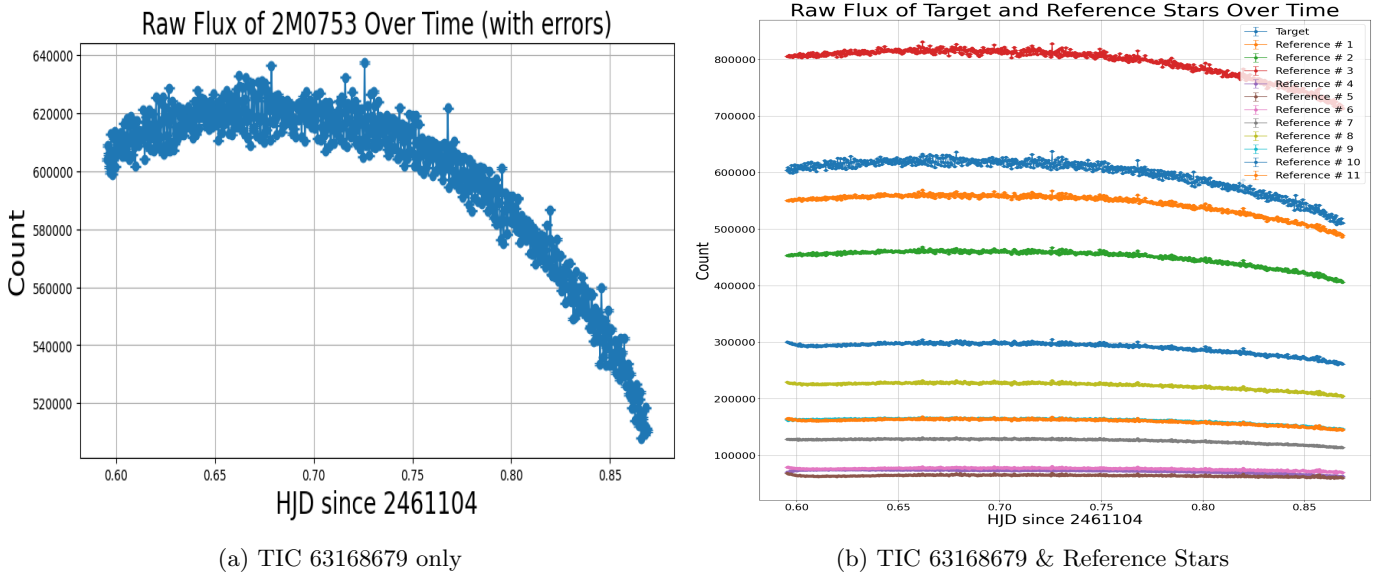


Figure 7: (Unnormalized) Light Curves of TIC 63168679 (Fig. 7a) and TIC 63168679 & reference stars (Fig. 7b.)

the hot subdwarf. Beyond that, based on Fig. 8b, our data appears to be pseudo-periodic, meaning that we can fit multiple sinusoidal curves with different amplitudes and frequencies to the light curve (see Fig. 9 for more details). This implies the importance of having the Nyquist frequency cutoff for the upper limit to our unevenly-spaced time values in order to probe for p-mode region only, as discussed in section 2.5.

3.3. Lomb-Scargle Periodogram

In order to further demonstrate the frequency aliasing mentioned in section 3.2, we plotted the frequency and power outputted by our LS routine when we inputted the time stamps, TIC 63168679's normalized flux, and the uncertainties in flux (see Fig. 10). Even though all the peaks from our sampling appear to be "real" since the probabilities of them not being noise are 1, they are simply aliasing signals from our photometric data. The different frequencies are from

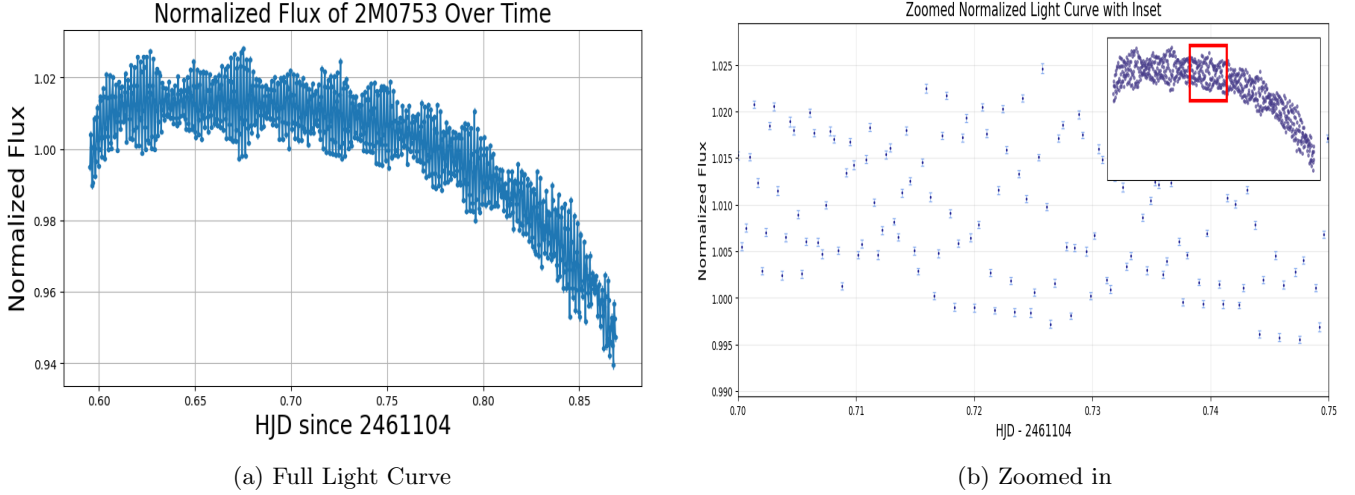


Figure 8: Normalized Light Curve of TIC 63168679 (2M0753).

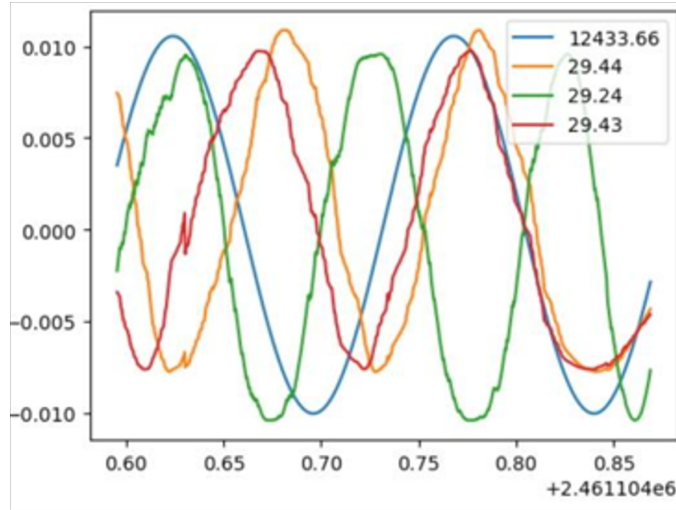


Figure 9: An illustration of frequency aliasing in our data by how we can fit multiple sinusoidal curves with many frequencies (colored lines, labeled numbers are frequencies in μHz) to a zoomed in portion of the normalized light curve.

the different sinusoidal lines being fitted, as shown in Fig. 9. Therefore, in order to remove this frequency aliasing and probe for p-mode frequencies, we had to adopt an upper frequency cutoff at the Nyquist frequency.

For our normalized photometry data, our LS routine calculated the Nyquist frequency to be $17016.675508590666 \mu\text{Hz}$, which is now the upper limit of the frequency grid used for LS sampling. As described in section 2.5, the lower limit was chosen to be $100 \mu\text{Hz}$. Additionally, based on the fact that the normalized flux appeared to evolve quadratically over time, we decided to adopt an extra step in our LS routine by removing slow trend from our normalized flux to the second order. Using the aforementioned frequency lower and upper limits, we calculated the 0.1% FAP level, which is marked in the new, Nyquist-limited LS periodogram (see Fig. 11). We also reported a table of frequencies of the pulsations we detected from TIC 63168679 (see Table. 3 for more details).

4. DISCUSSION

From our Nyquist-limited Lomb-Scargle periodogram, we detected possible pulsations with frequencies as seen in 3. The strongest frequency that we detected is $7130.16 \mu\text{Hz}$, which is less than 0.00121% away from the strongest frequency, $f_8 = 7130.24 \mu\text{Hz}$, reported in Table B.1 in Paper II. The other frequencies that we reported were also

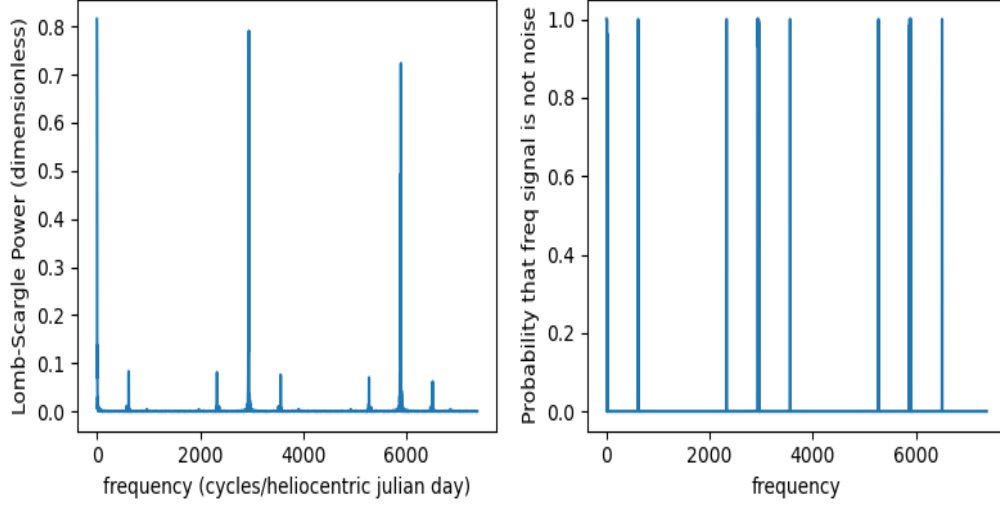


Figure 10: (Left): Raw Lomb-Scargle periodogram sampled over the entire frequency with our unevenly-spaced time stamps, TIC 63168679's normalized flux, and its uncertainty. (Right): This plot shows the probability of the pulsations in the left panel being "real" signal.

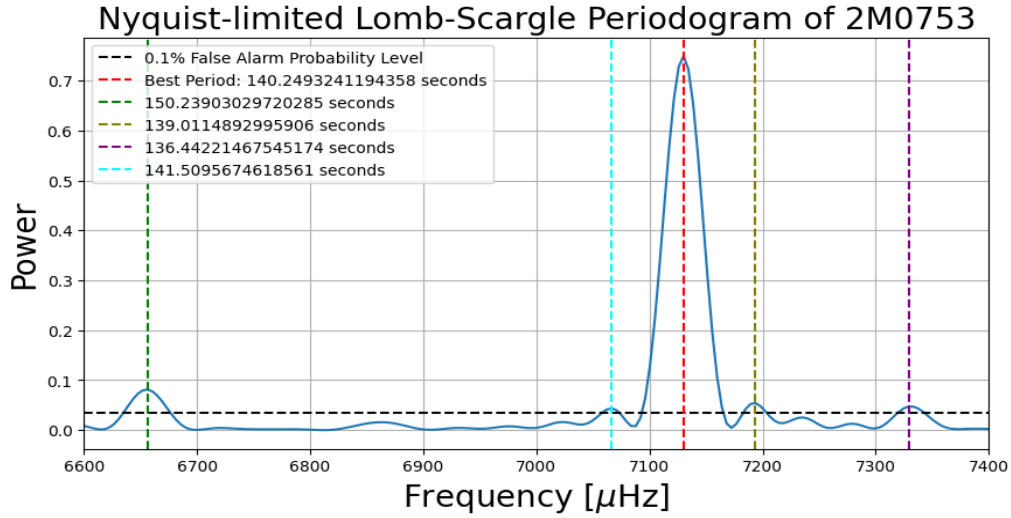


Figure 11: Lomb-Scargle periodogram with sampling limit from 100 μHz to Nyquist frequency of our data ($\sim 17016.68 \mu\text{Hz}$). The black horizontal line marks the 0.1% FAP level. The red vertical line marks the pulsation with the strongest signal, while the other vertical lines mark other possible pulsations stronger than the FAP level.

ID	Frequency [μHz]	Period [s]	Power
f_1	6656.06	150.24	0.08
f_2	7066.66	141.51	0.0432
f_3	7130.16	140.25	0.7484
f_4	7193.65	139.01	0.0547
f_5	7329.11	136.44	0.0471

Table 3: Pulsations we detected in our data of TIC 63168679 with their corresponding frequencies, periods, and power.

relatively close to the values reported in Paper II. However, we were not able to resolve the multiplets reported in Paper II. However, the widths of the peaks marked in Fig. 11 hint at the existence of these multiplets. Apart from that, one of the pulsations reported in Paper II, $f_6 = 6856.76 \mu\text{Hz}$ were not strong enough in our data to pass the detection threshold. Although this can be explained by the lower resolution of the 61-inch telescope compared to TESS and the shorter amount of observing time compared to Paper II, if we used the reasoning, then there would be another possible peak at $\sim 7220\text{--}7330 \mu\text{Hz}$. Therefore, either the reported frequency of $f_6 = 6856.76 \mu\text{Hz}$ in Paper II is a false alarm signal, or we detected a new pulsation around that range of frequencies.

Moreover, Paper II presented relationships between the period of the strongest pulsation with T_{eff} and $\log(g/cm s^{-2})$ in their Fig. 5 and Fig 6., which were imported as Fig. 12 and Fig. 13, respectively. Using our strongest pulsation with period $T_{max} = 140.25\text{s}$, the temperature and $\log g$ of TIC 63168679 are estimated to be $T_{eff} \approx 35,000 \text{ K}$ and $\log g \approx 5.8$. Because surface gravity $\log g$ depends on mass M and radius R of the star per Newton's gravitational law, and radius depends on luminosity and temperature of the star per Stefan-Boltzmann's law as follows,

$$\log g \sim \frac{M}{R^2} \sim \frac{M}{LT^{-4}} \sim \frac{MT^4}{L} \quad (5)$$

using solar values of $\log g_{\odot} \approx 4.44$, $T = T_{\odot} \approx 5800\text{K}$, we calculated the mass-to-luminosity of this star to be

$$\frac{M_{\star}}{L_{\star}} \approx 9.94 \times 10^{-4} \left(\frac{M_{\odot}}{L_{\odot}} \right) \quad (6)$$

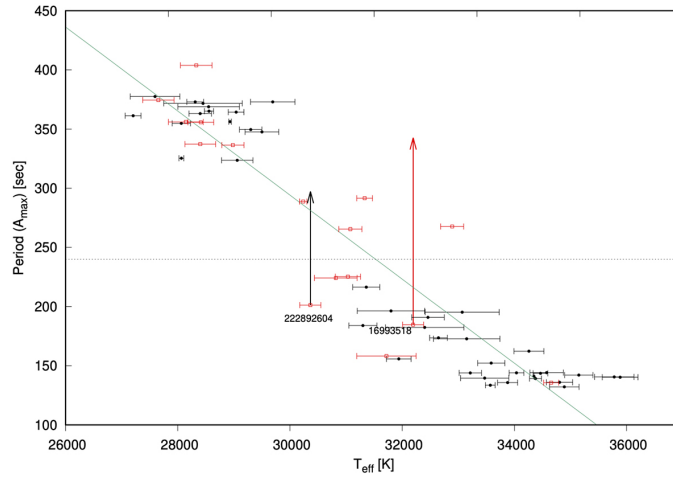


Figure 12: Period of the strongest pulsation as a function of T_{eff} . The sloped line represents a possible relationship between two quantities using their data. Imported from Paper II.

The extremely low mass-to-luminosity ratio indicate that this object must be very bright, thus agreeing with its O spectral type. In addition, Paper II also reports the distribution of hot pulsators in the $\log g$ - T_{eff} plane (see Fig. 14). Using the derived $T_{eff} \approx 35,000 \text{ K}$ and $\log g \approx 5.8$, TIC 63168679 would be on the track of a terminal-age extreme horizontal branch (TAEHB) star with a core mass of $0.47M_{\odot}$. Therefore, we concluded that its mass must be at least equal to it or greater, since we could not account for the mass of the outer layers.

Future implications of this project includes securing more observing time with the 61-inch facility and Mont4K CCD to continuously monitor the star's pulsations and confirm its possible multiplets, implied by the widths of our current peaks, which can help us derive its rotation period (anticipated to be 10.5 days in Paper II). In addition, we can also obtain the spectra of this star using the 90-inch facility that is also run by Steward Observatory, and we can utilize the classic method of fitting the spectral energy distribution (SED) to derive its temperature and confirm with our current result.

5. SUMMARY

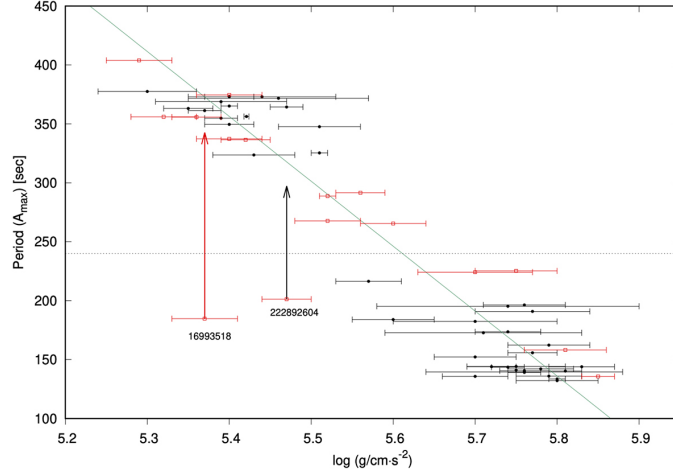


Figure 13: Same as Fig. 13, but with a function of $\log g$.

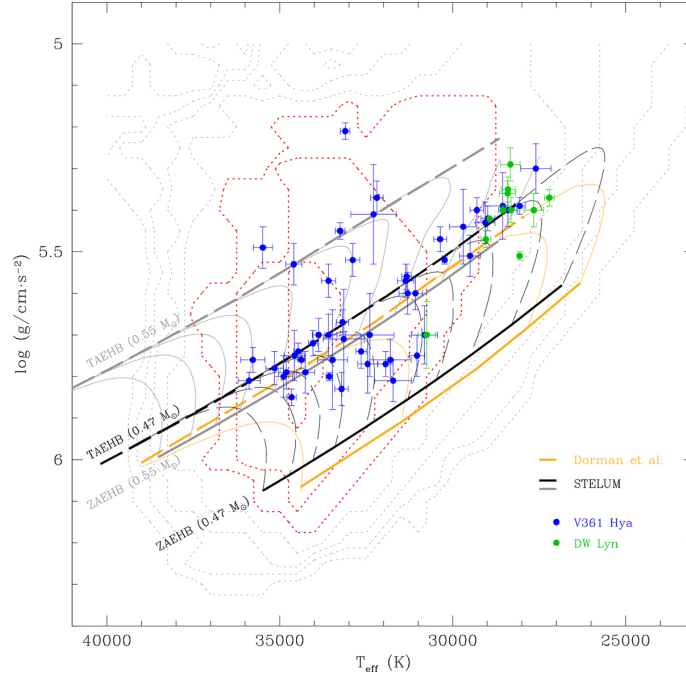


Figure 14: Distribution in the surface gravity ($\log g$) – effective temperature (T_{eff}) plane of the hot subdwarf stars with p-mode pulsations observed by TESS. The ZAEHB and TAEHB positions are indicated for an assumed core mass of $0.47 M_{\odot}$ (black curves) and $0.55 M_{\odot}$ (grey curves), illustrating how the EHB location depends on stellar mass. Imported from Paper II.

This work highlights the critical role of ground-based asteroseismology in validating space-mission data to understand the internal structures of evolved stars. By successfully confirming the pulsations of TIC 63168679 using 6-hour worth of ground-based observations, we demonstrate that high-precision photometry can accurately probe the evolutionary stages of hot subdwarfs. The analysis characterizes the target as a bright, O-type subdwarf nearing the end of its core helium-burning phase, contributing valuable data to the study of stellar formation theories such as common envelope evolution and binary mergers. Moving forward, we emphasize the need for extended monitoring to resolve suspected frequency multiplets and spectroscopic analysis to further refine the star’s rotation period and chemical composition.

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6. APPENDIX A: FINDER CHART USED DURING OBSERVING SESSION

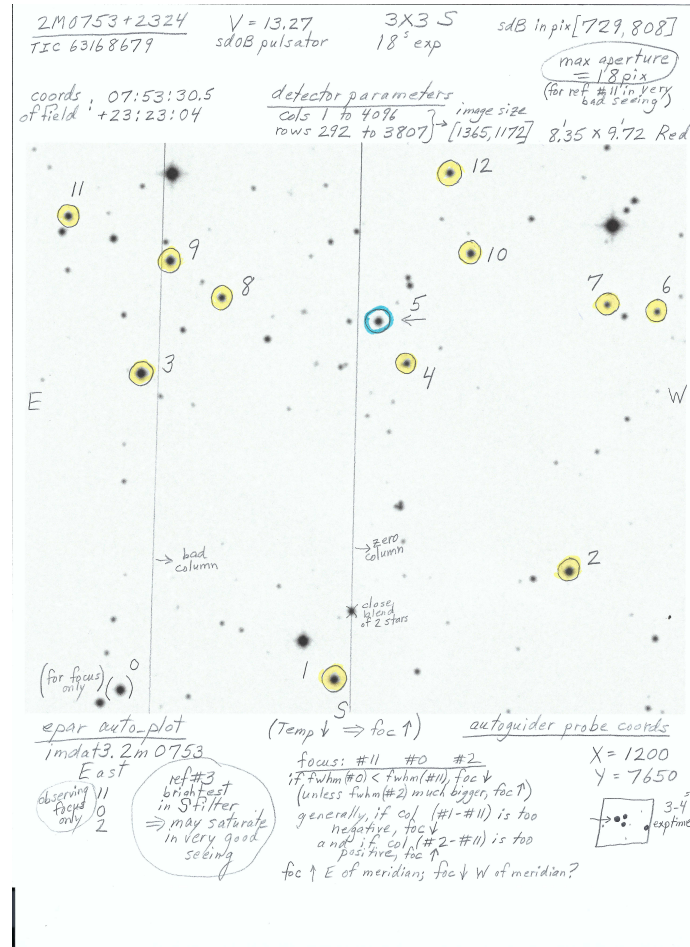


Figure 15: Finder chart with information about the target, the detector, and how our observing session was conducted, provided by Dr. Green. The coordinates at the top left were the actual coordinates of TIC 63168679 during the observing session. The approximate location of TIC 63168679 in an image is marked in blue, while the approximate positions of the reference stars are marked in yellow circles. The vertical lines mark the bad columns on the CCD.

7. APPENDIX B: CALIBRATION FRAMES

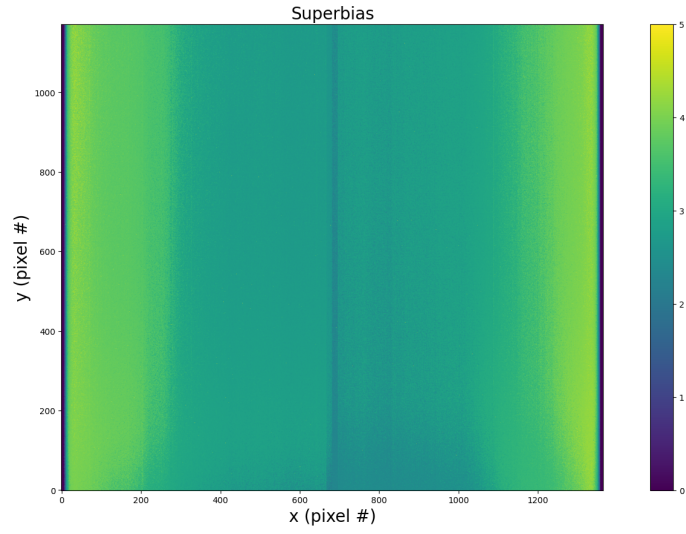


Figure 16: Superbias frame to account for electron read noise of the CCD. The central vertical line is one of three bad columns defined in Fig. 15.

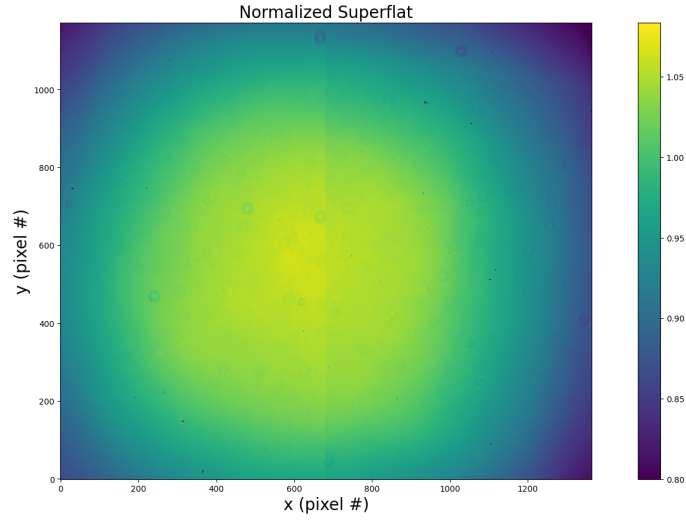


Figure 17: Normalized, superflat frame to account for slight vignettes at the corners of the images from using the S-filter.